Design and Simulation of

WCSPH Simulation of Liquid Sloshing in a Pivoted Rectangular Tank: Grid Convergence, Hydrostatic Verification and Two-Way Coupled Active Control

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Introduction

Sloshing in a road tanker

Sloshing

Carry a full cup of coffee and stop suddenly: the coffee keeps going, climbs the wall and spills. This wave motion of a free liquid surface inside a moving container is called sloshing.

In a partly filled road tanker, the same thing happens with tonnes of liquid, and the liquid pushes back on the tank and the truck.

Braking

The liquid surges forward and the tank pitches; the surge pushes the truck forward as it tries to stop

Acceleration and bumps

The liquid surges backward and keeps rocking after the vehicle motion has ended

Safety risk

The moving liquid shifts the load at the worst moment and lowers stability (Zheng et al., 2017)

Today's fix: baffles

Fixed plates inside the tank add weight and cannot adapt; our project studies an active, motor-driven alternative

Main application: road tanker compartment

Our tank is a 1:5 scale model of a tanker compartment pitching under braking

Why a road tanker

When a tanker brakes or accelerates, the tank pitches and the liquid surges along its length.

This is the same motion as our tank tilting about its pivot.

An active tilting cradle under the tank, driven by a motor, could cancel the surge: this is what our controller does.

Why Froude scaling: gravity is the restoring force in sloshing; surface tension (Bo ≈ 22,000) and viscosity (Re ≈ 7.7×10^4) are negligible at this size.

Table 1. Our tank as a 1:5 scale model (Froude scaling, same proportions)

Quantity	Our tank (model)	Tanker compartment
Length (sloshing direction)	0.40 m	2.0 m
Width	0.25 m	1.25 m
Height	0.30 m	1.5 m
Fill depth	0.20 m (h/L = 0.5)	1.0 m (h/L = 0.5)
Sloshing frequency	1.34 Hz	0.60 Hz
Time scale	1 s	2.24 s

Lengths × 5; frequencies ÷ √5; times × √5 (from Eq. 9 of our report)

The liquid sloshes within one compartment (bulkheads separate them), so we scale to a 2 m compartment, not the ~10 m tanker.

Problem Statement

From Section 1.3 of our report

A rigid rectangular tank, partly filled with water, rotates about one horizontal pivot at its base under a motor torque: a 1:5 scale model of a road tanker compartment pitching under braking. Simple equivalent models of the water are used to design controllers for such a tank, but their accuracy for this motion is unknown, and so is the accuracy of the detailed simulation used to check them.

1. **How accurate is the detailed simulation?**
   Does the SPH solution converge to a known exact answer as the particles are made smaller?

2. **How good is the simple model?**
   Over a realistic, controller-generated tilt, how well does the equivalent pendulum match the detailed simulation?

3. **Do the controllers work on real water?**
   With the water pushing back on the tank (two-way coupling), how do controllers designed on the simple model perform?

Literature Review

Table 2. Equivalent models, sloshing theory and tank experiments

Sr.	Year	Author	Title	Relevance to our work
1	1963	G. W. Housner [1]	The dynamic behavior of water tanks	Split the water into an impulsive mass and a convective mass on a spring or pendulum. Basis of our pendulum model.
2	2006	ACI Committee 350 [2]	Seismic Design of Liquid-Containing Concrete Structures (ACI 350.3-06)	Closed-form formulas for the two masses, their heights and the first sloshing frequency, used in our code.
3	1966	H. N. Abramson (ed.) [3]	The Dynamic Behavior of Liquids in Moving Containers (NASA SP-106)	Classic study of sloshing in space vehicles; shows why sloshing control matters.
4	2005	R. A. Ibrahim [4]	Liquid Sloshing Dynamics: Theory and Applications	Linear potential theory of sloshing; gives the exact first-mode frequency we use as a check.
5	2005	H. Akyıldız, E. Ünal [14]	Experimental investigation of pressure distribution on a rectangular tank due to liquid sloshing	Measured wall pressures in a pitching rectangular tank at several fill depths and amplitudes.
6	2024	K. Liu et al. [15]	Experimental study on the sloshing of a rectangular tank under pitch excitations	Strong nonlinearity near resonance under pitch excitation; confirms that pitching tanks need careful study.

Literature Review

Table 3. SPH for sloshing, verification and sloshing control

Sr.	Year	Author	Title	Relevance to our work
7	2006	A. Souto-Iglesias et al. [7]	Liquid moment amplitude assessment in sloshing type problems with SPH	Compared SPH with rolling-tank experiments; supports SPH as a sloshing tool.
8	2011	A. Rafiee, F. Pistani, K. Thiagarajan [9]	Study of liquid sloshing: numerical and experimental approach	SPH agreed well with laboratory pressures and free-surface shapes.
9	2012	S. Adami, X. Y. Hu, N. A. Adams [21]	A generalized wall boundary condition for SPH	The wall method (ADAMI) used in our SPH tank.
10	2008	I. B. Celik et al. [23]	Procedure for estimation and reporting of uncertainty due to discretization in CFD	The grid convergence index (GCI) method used in our resolution study.
11	1997	J. T. Feddema et al. [24]	Control for slosh-free motion of an open container	Classical model-based slosh control, built on equivalent models.
12	2025	M. Ye et al. [30]	Sloshing suppression with a controlled elastic baffle via deep reinforcement learning and SPH	Closest recent work: learning control with SPH, but it moves a baffle, not the tank itself.

Research Gap

What the literature does not cover, and how this project addresses it

Table 4. Gaps in the reviewed literature

Gap in reviewed literature	How this project addresses it
SPH sloshing studies use sideways or roll motion about the tank centre, at one or two resolutions [7, 8, 9]	Pitch about a pivot at the base, at four particle sizes, with a GCI error estimate
Controllers are designed on equivalent models whose validity for this motion is assumed, not checked [24, 25, 26]	The pendulum model is checked against verified SPH over a controller-generated motion
SPH accuracy is usually judged against experiments, without a separate numerical error estimate	SPH is verified against an exact hydrostatic answer: 3.1% error, GCI 6.4%
Learning-based sloshing control with SPH moves baffles inside the tank [29, 30]	Our actuator is the tank's own pivot motor, with the water pushing back (two-way coupling)

Objectives

From Section 1.3.1 of our report

1. **Build the model**
   Model the pivoted tank in SPH (Chrono::FSI) with one-way (prescribed motion) and two-way (torque-driven) coupling to the rigid tank

2. **Measure the numerical error**
   Run a four-level particle-size study and estimate the grid convergence index (GCI)

3. **Check against exact answers**
   Compare with the hydrostatic water shift and the first sloshing frequency from potential theory

4. **Test the simple model**
   Measure how well the Housner pendulum matches SPH along a controller-generated motion

5. **Compare controllers**
   Test PID, LQR and a learning controller (PPO) in the two-way coupled SPH tank

Methodology

Six steps, from design to validation

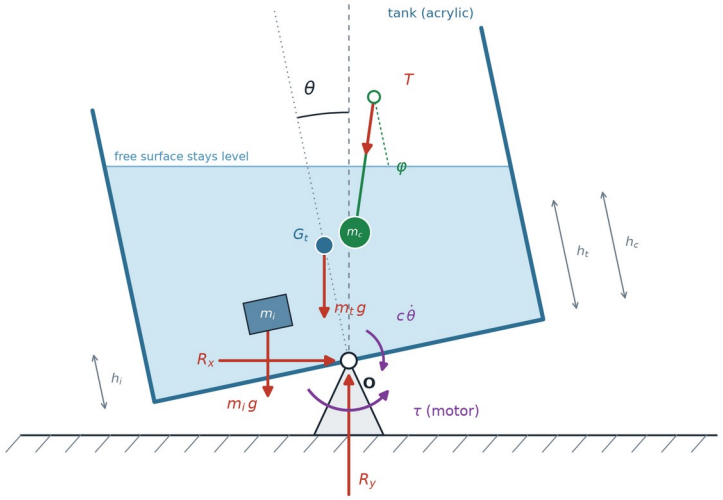
1 → 2 → 3 → 4 → 5 → 6

1	2	3	4	5	6
Define	**Model**	**Train**	**Simulate**	**Verify**	**Compare**
Application (road tanker), tank design and requirements	Tank, pivot and motor; pendulum water model; RL environment	PPO, PID and LQR on the fast model	SPH water on a GPU at four particle sizes	Convergence and exact-answer checks	Controllers in the two-way SPH tank
DESIGN		**ANALYSE**		**VALIDATE**	

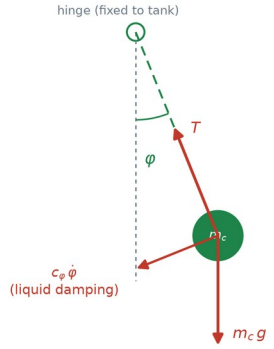
Free-body diagram

The tank on its single pivot O, with the water replaced by its equivalent model

(a) Tank + liquid on a single pivot O



(b) Convective mass m_c isolated



Values used in the model (water, 400 x 250 x 300 mm acrylic tank)

Tank mass	m_t	3.47 kg	Tank inertia about O	I_O	0.156 kg m^2
Tank CoM above O	h_t	119 mm	Liquid fill height	H	200 mm
Impulsive mass	m_i	10.85 kg	Impulsive height	h_i	62.5 mm
Convective mass	m_c	9.70 kg	Convective height	h_c	117 mm
Pendulum length	L	138 mm	Slosh frequency	f_n	1.34 Hz
Liquid damping ratio	ζ	0.5 %	Pivot damping	c	0.02 N m s

Equations of motion (small angles)

Tank about O: $I_O\ddot{\theta} = \tau - c\dot{\theta} + (m_t h_t + m_i h_i)\, g\,\theta + M_T$

m_c pendulum: $\ddot{\varphi} = -\frac{g}{L}\varphi - 2\zeta\omega_n\dot{\varphi} + (\text{tank forcing})$

Slosh energy: $E = \frac{1}{2}m_c(L\dot{\varphi})^2 + m_c gL(1-\cos\varphi)$

Natural frequency: $\omega_n^2 = g/L$

M_T = moment of the rod tension T about O. The $+g\,\theta$ term is destabilising: G_t sits above O, so the tank behaves like an inverted pendulum.
Simulator simplifications: the fast training model omits the gravity term; its pendulum is forced by the tank's angular velocity $\dot{\theta}$ only.
The real-fluid (SPH) and Chrono tank simulations include gravity.

Fig. 1. Free-body diagram: motor torque, pivot damping and reactions, tank weight, and the Housner equivalent of the water

Motor torque τ

applied at the pivot to tilt the tank

Tank weight

acts 119 mm above the pivot: the tank is an inverted pendulum

Water

impulsive mass fixed to the tank, convective mass on a pendulum

Choosing the tank size and values

Every value has a stated reason (second review)

Table 5. Design choices and their basis

Quantity	Value	Why this value
Tank length L	0.40 m	Bench-top lab scale (tests use tens of cm to ~1 m [4, 5, 9]); a 1:5 Froude model of a 2 m tanker compartment; cheap to build, fill and film
Width / height	0.25 m / 0.30 m	Narrower than long, so the first mode lies along the tilt axis; 100 mm freeboard (the surface rises only ~35 mm at 10°)
Wall	6 mm cast acrylic	Standard sheet; transparent, so the free surface can be seen and filmed
Fill depth h	0.20 m (h/L = 0.5)	Inside the valid range of the ACI 350.3 formulas; water splits almost evenly into impulsive and sloshing parts
Sloshing damping	0.5% of critical	Standard value for a smooth-walled water tank (Sivý et al. [13])
Resulting frequency	1.34 Hz	Easy to film and follow with a modest motor; equals ≈0.6 Hz in a 2 m tanker compartment
Liquid	Water, 1000 kg/m^3	Simplest to begin with: safe, well-known properties, the standard reference liquid. Petrol gives the same 1.34 Hz, with ~25% less mass

Modelling the water as a pendulum

The Housner equivalent model (ACI 350.3-06)

The idea

When the tank moves, the water near the bottom moves with it like a solid; only the water near the surface sloshes. Housner (1963) therefore split the water into two masses:

Impulsive mass: fixed rigidly to the tank

Convective mass: hangs on a pendulum and swings

$\omega^2 = (3.16\ g / L) \tanh(3.16\ h / L), \quad L_p = g / \omega^2$

A swinging pendulum is far cheaper to compute than real water.

10.85 kg
impulsive mass (moves with the tank)

9.70 kg
convective mass (sloshes)

0.138 m
equivalent pendulum length

1.343 Hz
first sloshing frequency

Note: in our code these formulas use the outer tank size (20.0 kg of water). The two masses add up to 2.7% more than that, which is a known feature of these curve-fitted formulas.

System architecture

Like a modular test rig: parts can be swapped

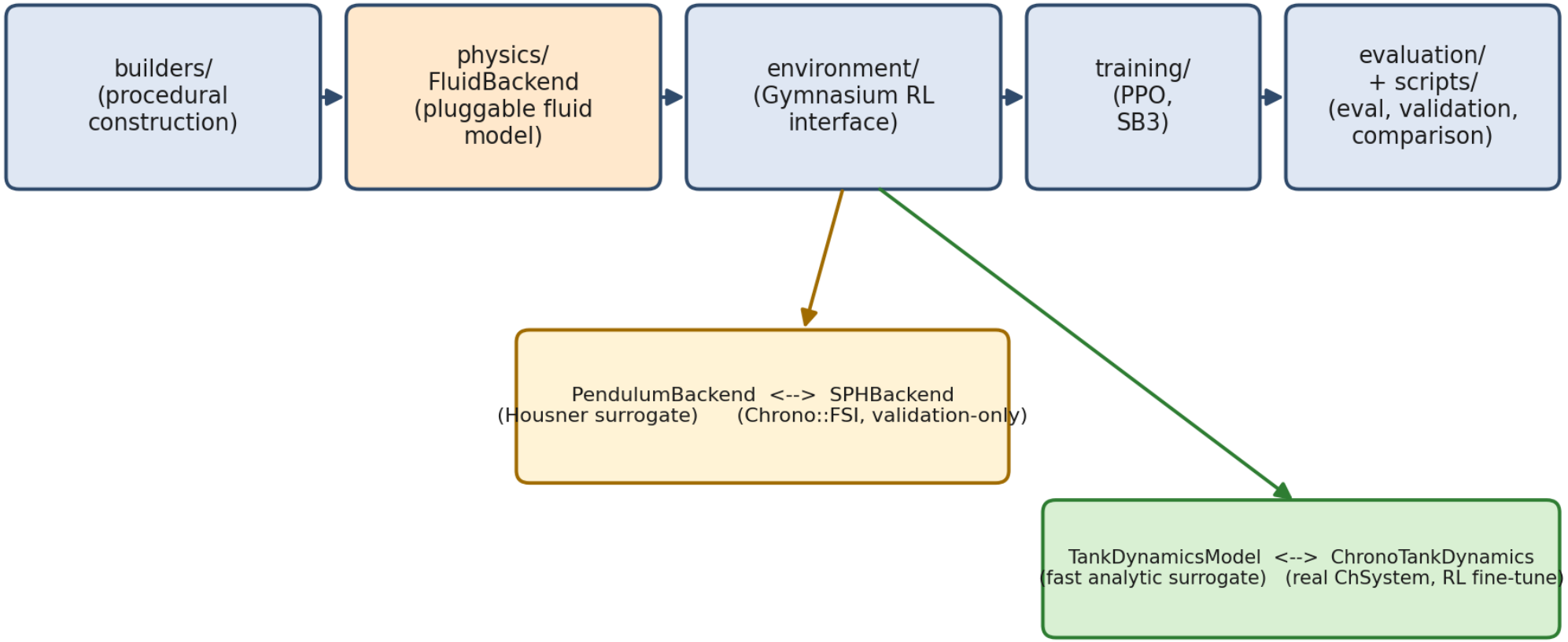


Fig. 2. Software architecture: the water model (pendulum ↔ SPH particles) and the tank model (fast ↔ full multibody) can each be swapped without changing anything else

Checking the model by hand

Hand calculations in Excel compared with the simulation

Table 6. Hand calculation vs simulation

Quantity	Hand calculation	Simulation	Difference
Tank mass, centre of mass, inertia	Plate formulas + parallel axis	Same values in the model	0.00%
Housner masses and pendulum length	ACI 350.3-06 formulas	Same values in the model	0.00%
Pendulum sloshing frequency	1.343 Hz	1.30 Hz (measured from the motion)	−3.2%
Sloshing frequency, detailed water	1.364 Hz (potential theory)	1.422 Hz (SPH)	+4.2%
Controller design check (LQR)	8.44 rad/s (Housner)	8.44 rad/s (closed-loop poles)	0.04%
Water shift at a held tilt of −10.16°	−11.24 mm (hydrostatic)	−10.89 mm (SPH, 2 mm)	−3.1%
Froude number	Fr = 0.054 (gentle, linear sloshing)	Same value	—
Reynolds number	Re ≈ 7.7×10^4	Same value	—

Open the workbook: **Calculation_vs_Simulation.xlsx**

Is the slosh-energy measure valid?

Checked after the second review

What we found

- The reported slosh energy was kinetic only: $E = \frac{1}{2} m v^2$
- For the pendulum this leaves out the potential term $m_c g L (1 - \cos \theta)$
- Measured: the potential term is 51–66% of the sloshing mode's energy, so kinetic-only under-reports it by about half

What we did

- Pendulum model: now reports kinetic + potential energy
- Ranking unchanged: total-energy settling PPO 1.20 s, LQR 3.72 s
- SPH: reported as bulk kinetic energy only; its height-based potential term is dominated by a numerical settling drift, so it is not used
- Training reward kept as before: driving kinetic energy to zero still drives total energy to zero

Why the tank needs active control

The tank is an inverted pendulum, and the water adds large forces

Centre of mass 119 mm above the pivot

Like a pencil balanced on its tip: level is unstable, and any small tilt grows on its own.

Measured: a tilt of 0.0001 rad grew to 0.161 rad in 0.5 s

Free-swing test with no motor, using the detailed water simulation.

The water's torque is about 7× the tank's own

Peak water torque 4.78 N·m against 0.70 N·m from the tank's inertia and weight combined.

Therefore the motor must keep correcting the tilt while the water pushes back on the tank. Doing this without making the water slosh even more is the control problem.

What a controller does

Measure → decide → act, fifty times a second

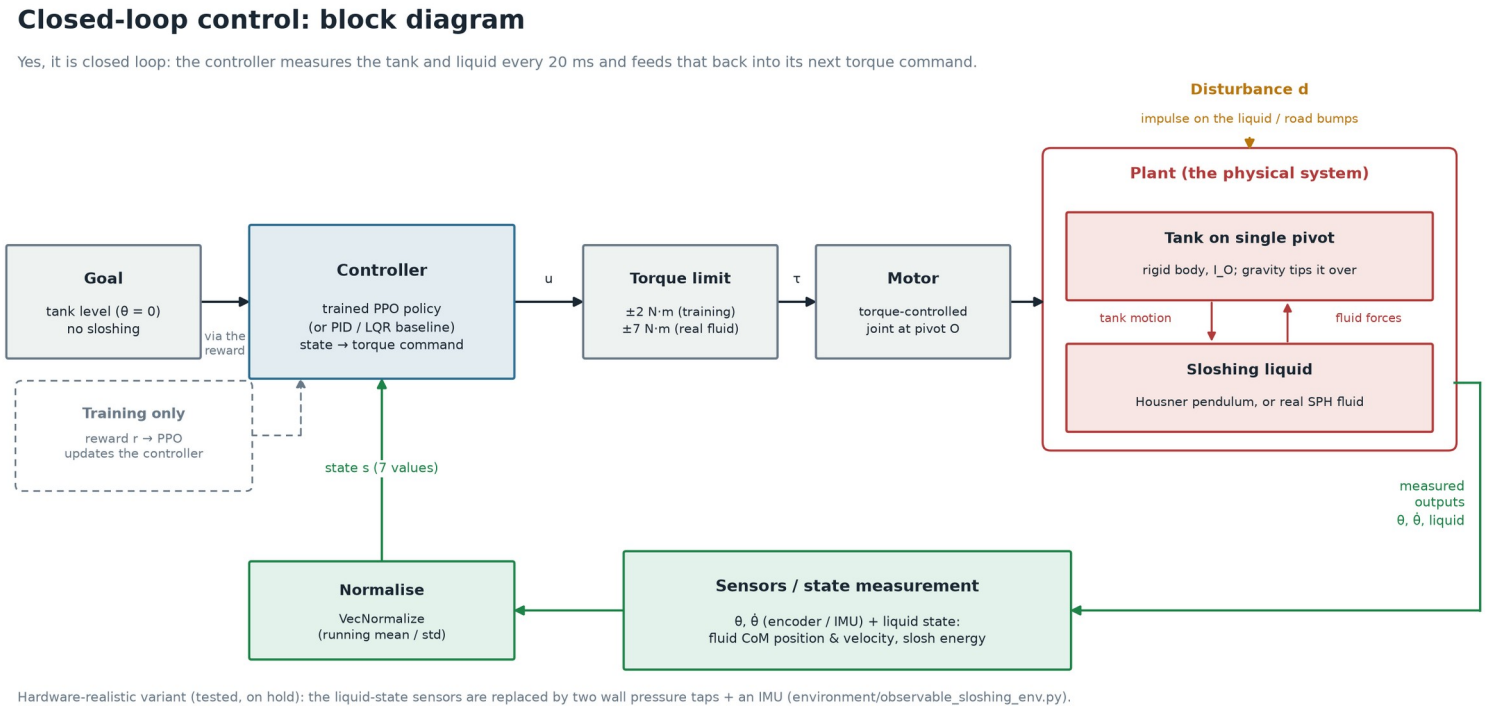


Fig. 3. Closed-loop block diagram

Closed loop

Like a car's cruise control: it measures, corrects and measures again

Every 20 ms

A new motor torque is chosen, 50 times per second

What it sees

Tank angle and speed, and the water's position, speed and energy

The controllers, side by side

Explained for a non-specialist

Table 7. Controller comparison

	PID	LQR	PPO (learned)
Everyday picture (carrying soup)	Keep the bowl level and hope	Work out the ideal tilt from a formula	An experienced waiter who just knows
What it looks at	Tank angle and its rate	Tank angle and rate, water swing angle and rate	7 numbers: tank, water position and velocity, slosh energy
How the rule is made	Hand-tuned gains: Kp = 14, Ki = 1.5, Kd = 4.5	Solved once from the pendulum equations (Riccati)	Trained by trial and error, 500,000 steps
Fast model: time to calm the water	Never reacts	3.4 s	1.2 s
Real SPH water	Stays in control; most sloshing left	Loses control in 0.7 s	15× less sloshing than PID; in control ~5 s

Why the motor torque is bounded

A real motor has a maximum torque; the limits were measured, not guessed

Table 8. Torque-limit sweep on the real (SPH) closed loop

Limit (N·m)	Survival (steps)	What happens
1.0	~60	Too weak: tank runs away
2.0	~40–50	Marginal, keeps oscillating
4.0	~100–125	Good while it lasts
5.0–6.0	~150	Good while it lasts
7.0	**~200–300**	**Best region found**
8.0	~200–210	Good, but shorter
10.0	~90–100	Too strong: over-corrects

Why a bound is needed

Every real motor saturates; without a cap the controller could ask for torque no motor can give.

Fast model, ±2 N·m: PPO's largest torque was only 0.08–0.19 N·m, so the bound never shapes the result.

Real water, ±7 N·m: above the water's 4.78 N·m push-back, and the best value in the sweep. A small industrial servo size.

All three controllers get the same limit.

How the controller learns

Trial and error, guided by a score

1. **Try**
   Run the tank for 10 s with a random water kick, choosing a torque every 20 ms

2. **Score**
   Lose points for sloshing energy, motor effort and tilt

3. **Learn**
   Adjust the controller so better actions become more likely

4. **Repeat**
   500,000 steps, 8 virtual rigs at once: about 3–5 minutes

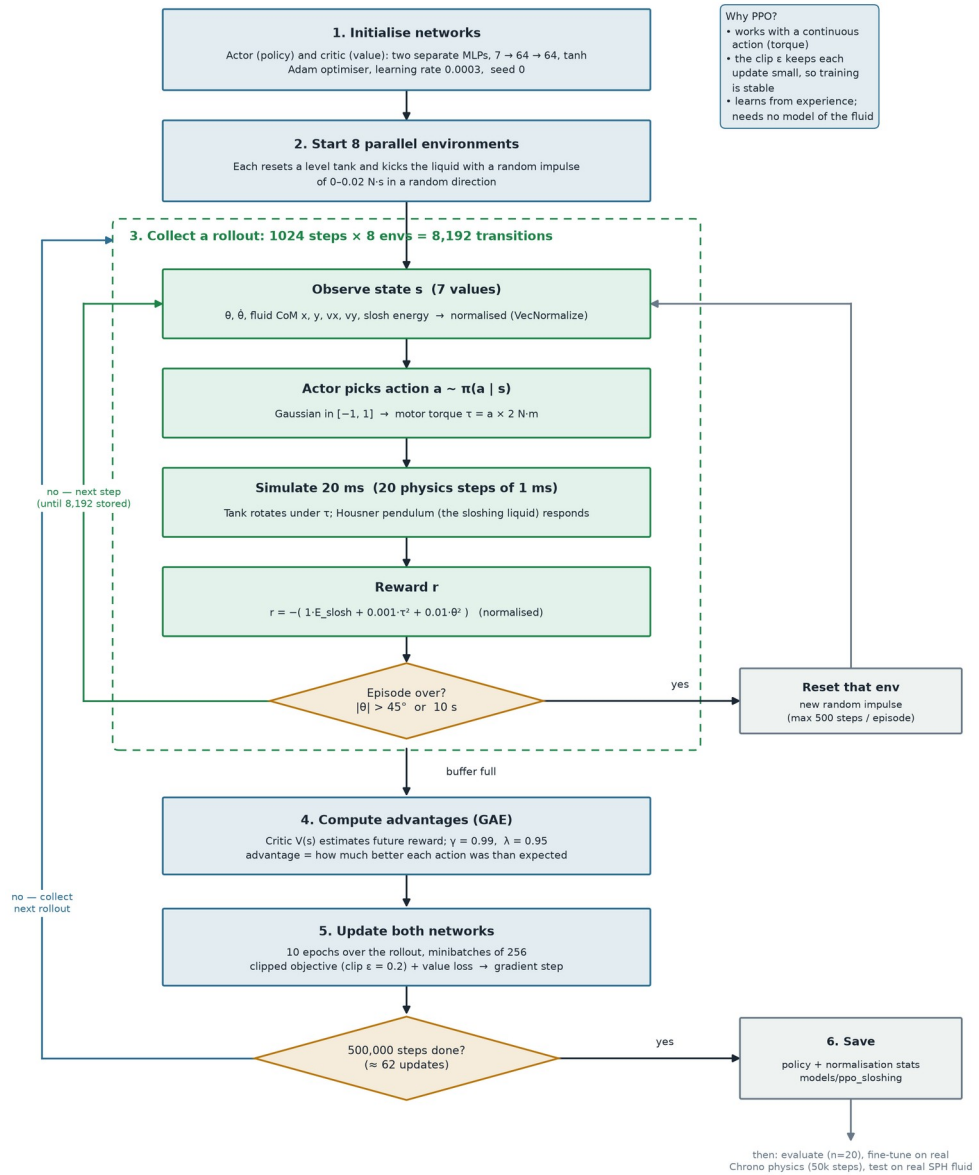


Fig. 4. PPO training flowchart (right)

Result on the virtual rig

All controllers, same water kick, same 10 s run

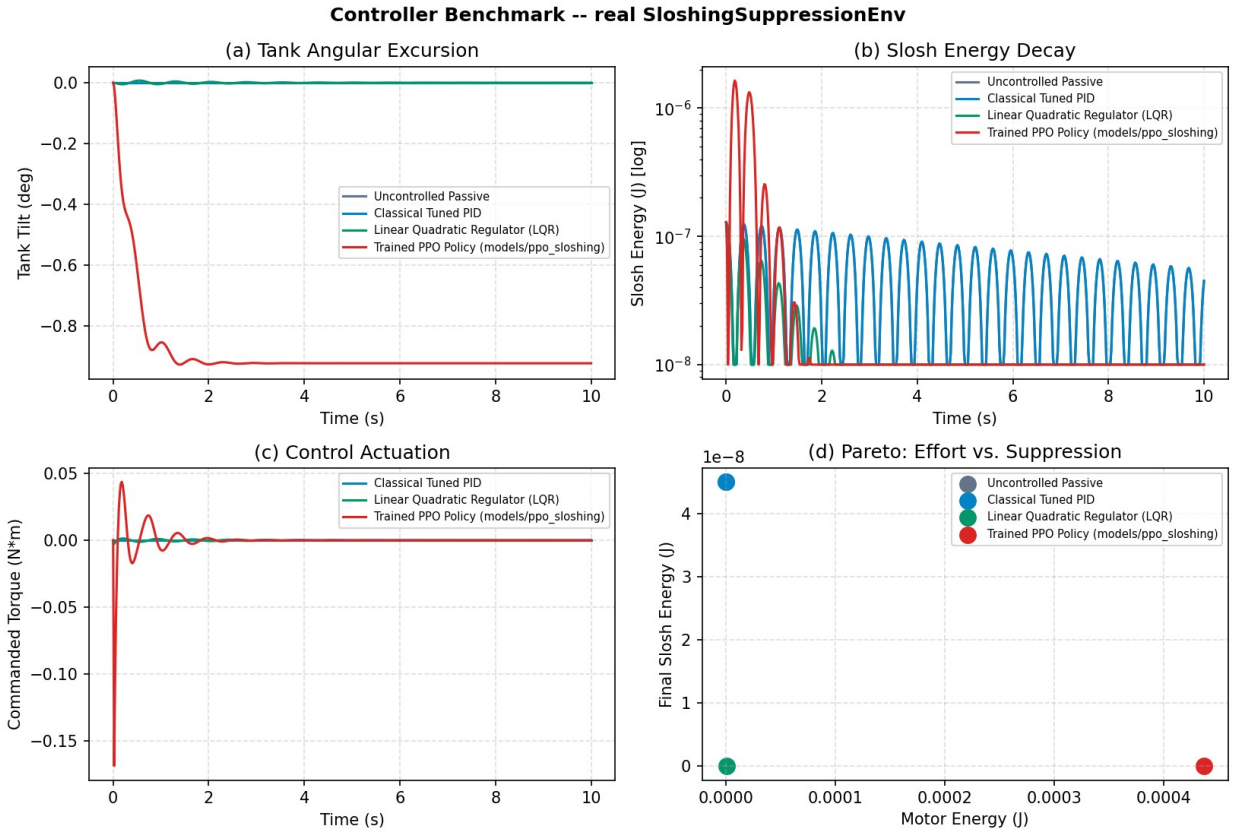


Fig. 5. Controller benchmark on the fast model

1.24 s

PPO calms the water fastest; LQR takes 3.42 s

PID does nothing

The kick hits the water, not the tank, so an angle-only PID sees no error at all

–0.9° tilt

PPO leaves the tank slightly tilted, because its score cares far more about sloshing than tilt

From the fast model to full physics

The controller had to be re-tuned for the full model

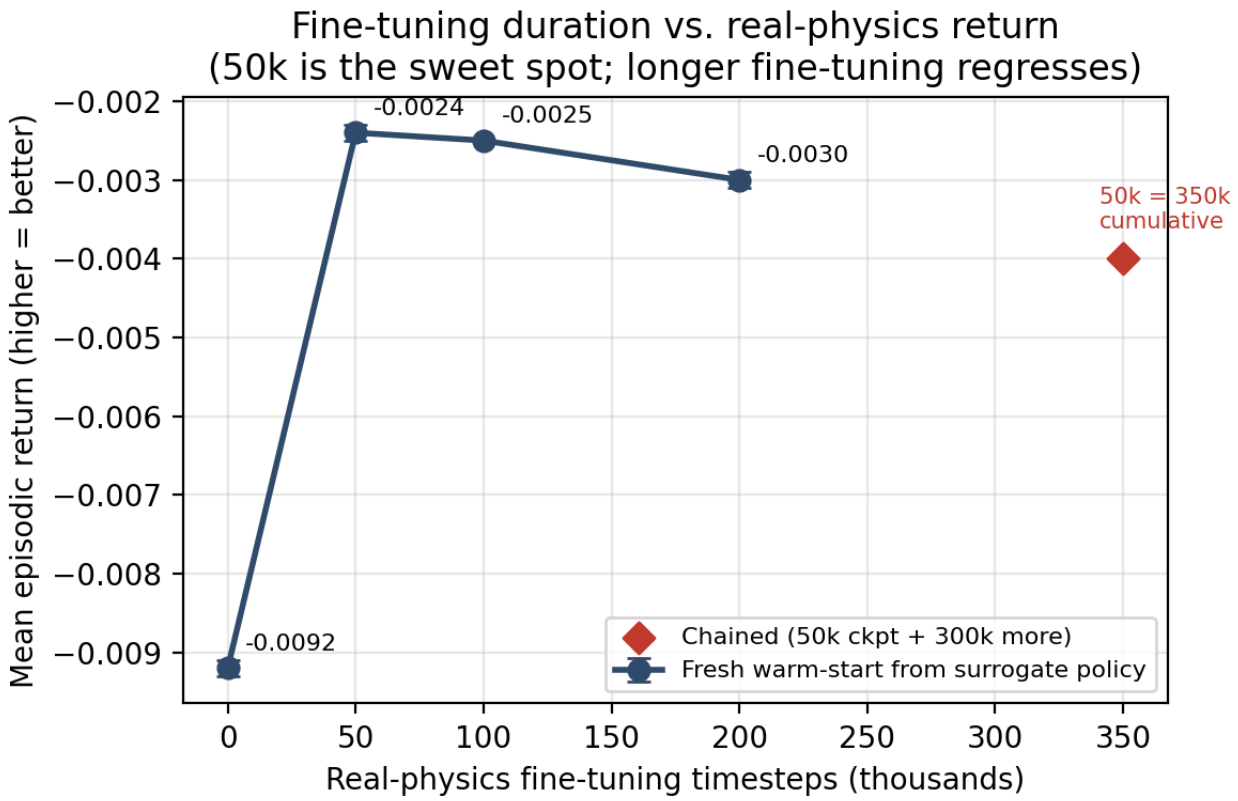


Fig. 6. Score on the full multibody model against extra training length

~11.5× more effort

Needed on the full multibody tank (friction and contact the fast model leaves out)

50,000 extra training steps

On the full model made it ~3.8× better

More was worse

100k, 200k and 350k steps all did worse, so 50k was kept

Simulating real water with particles

Smoothed Particle Hydrodynamics (SPH), on a graphics card

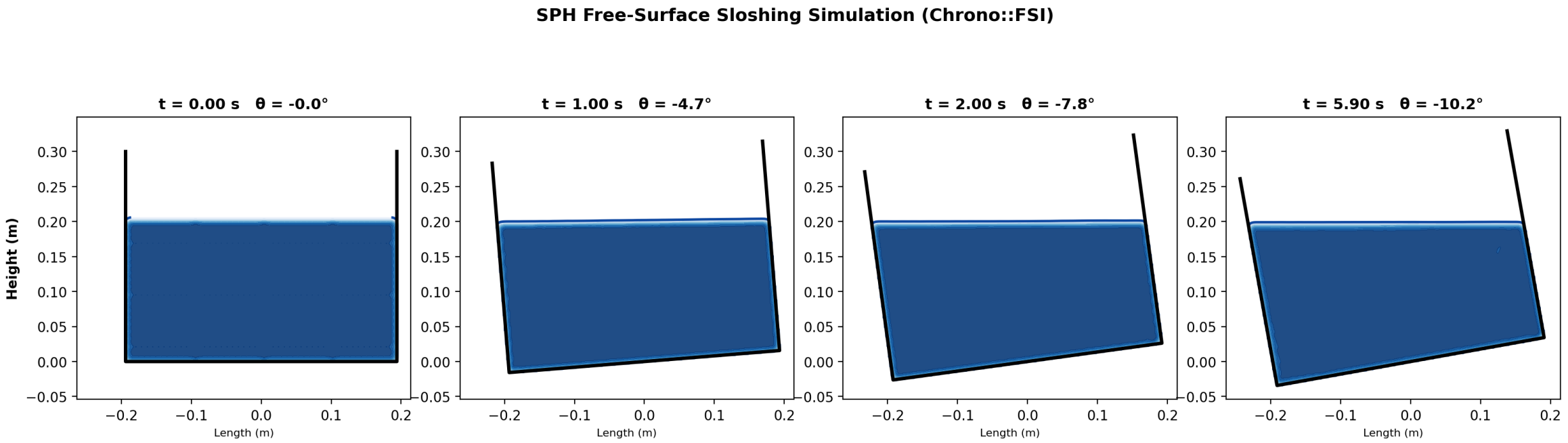


Fig. 7. SPH water at 2 mm particle size (about 2.26 million particles): the surface stays level while the tank tilts around it

Check 1: convergence

A grid-independence study, using particle size instead of mesh size

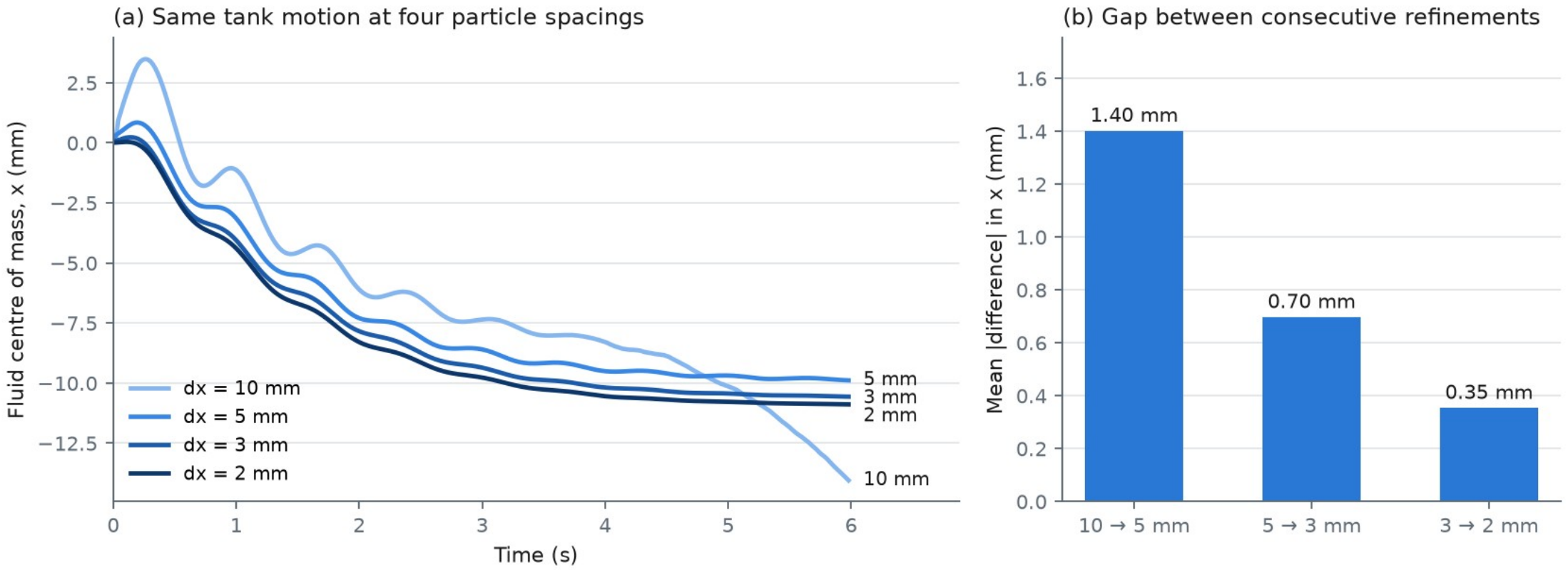


Fig. 8. Same tank motion at four particle sizes; the change between sizes keeps getting smaller (1.40 → 0.70 → 0.35 mm)

Check 2: exact answer

When the tank is tilted and held, the water must settle level

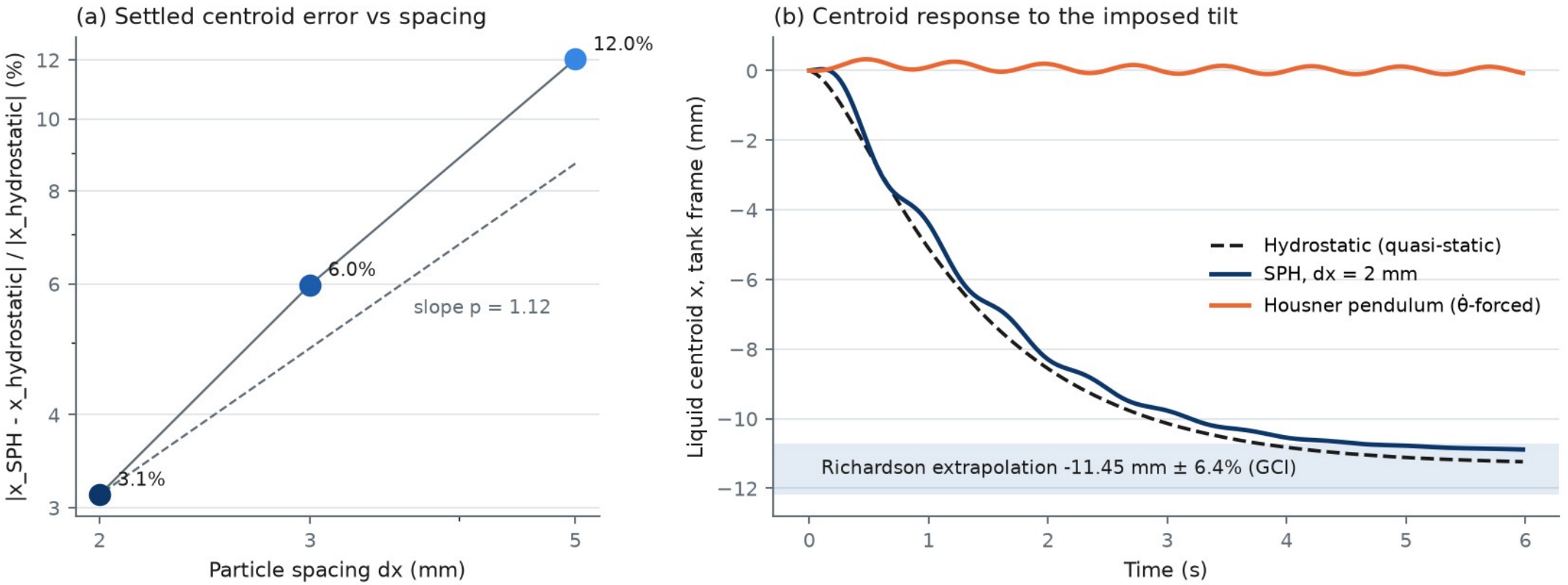


Fig. 9. (a) Error against the exact hydrostatic value falls to 3.1% at 2 mm; (b) SPH follows the exact answer (correlation 0.999), the pendulum does not (0.48)

Check 3: sloshing frequency

Tilt the tank, hold it, and let the water oscillate freely

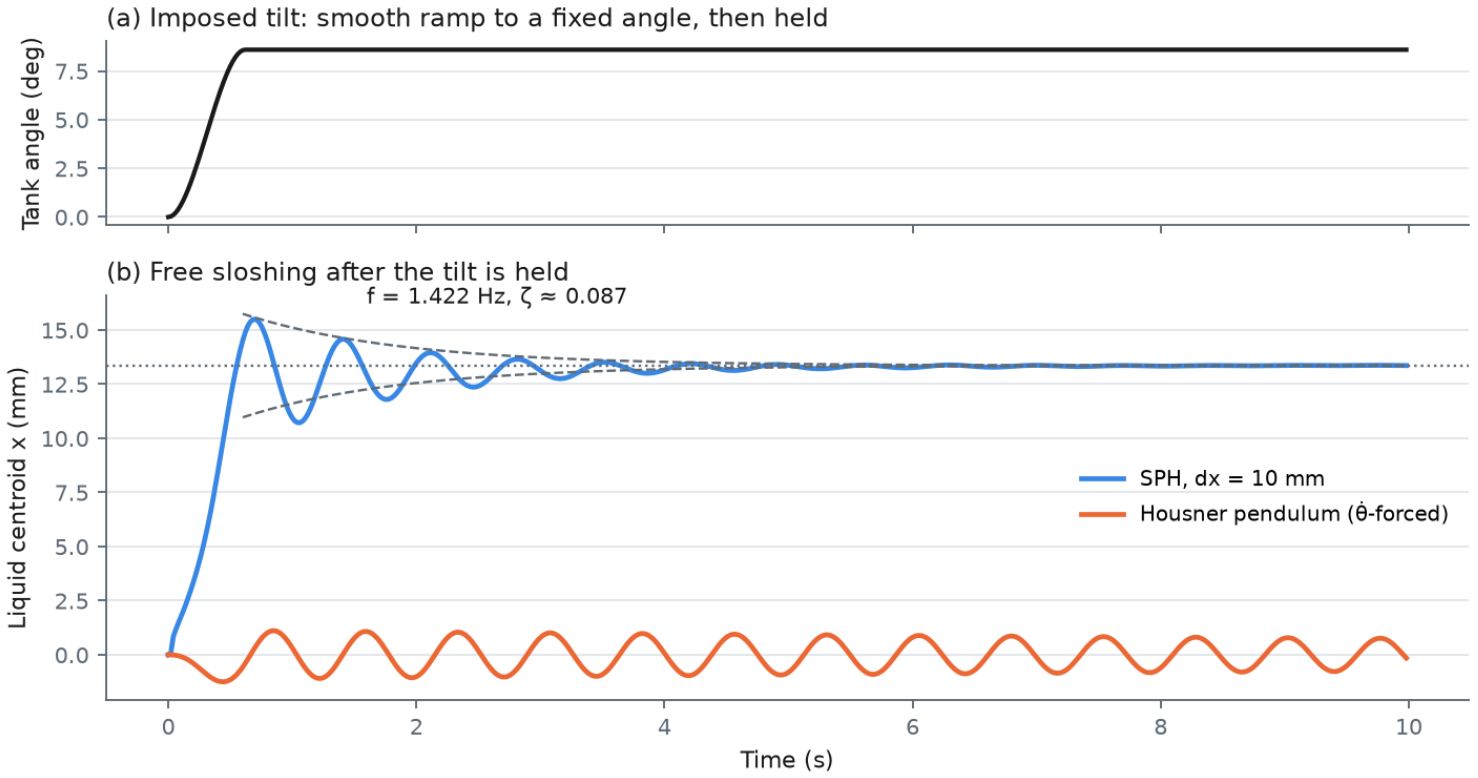


Fig. 10. Free sloshing after a held tilt of 8.59° (10 mm particles)

1.422 Hz

Measured in the SPH water (+4.2% vs potential theory, +5.9% vs Housner)

ζ ≈ 0.087

The SPH wave decays ~17× faster than in real water (ζ ≈ 0.005), mostly because of numerical damping at 10 mm

Different shape

SPH water shifts downhill and stays there; the pendulum oscillates about zero

Two-way coupled control

Single 300-step (6 s) runs in the torque-driven SPH tank

Table 9. Two-way coupled control results (report Table 8)

Controller	Steps before trip	Final energy (J)	Max. torque (N·m)	Motor energy (J)
PID (tank angle)	300 (no trip)	0.0157	4.26	1.12
LQR (pendulum model)	34	0.587	3.62	—
PPO (best checkpoint, 514560)	252	0.0010	4.69	1.19

~15× less

sloshing energy than PID with the learning controller

~5 s

how long the learning controller stays in control (≈11 s at 2 m tanker scale)

LQR fails

its gains come from the pendulum model, which misses re-levelling

Interactive simulations

Click to open our live simulations in a web browser

3D Sloshing Lab

- Recorded: replay of the real SPH water in the tilting tank
- Live: drive the tank yourself with the fast model
- Road: the tank on a truck bed over a bumpy road
- Resolution: compare the four particle sizes

Click to open: 3D Sloshing Lab

Opens in any web browser; works without internet

Interactive Project Report

- Live demo: set the motor torque, press “Kick it”, and try to calm the water
- Validation of the fluid model against real SPH
- Closed-loop results on real fluid, every key number
- Answers to the second review

Click to open: Interactive Project Report

Opens in any web browser; works without internet

Challenges

Engineering problems we hit, and how we solved them

Table 10. Challenges and solutions

Challenge	Cause (in plain words)	Solution
No CUDA graphics card on the laptop	The SPH solver runs only on NVIDIA GPUs	Rehearsed on Colab, then built the solver on an RTX 5060 and an RTX 3090
Fluid solver crashed immediately	Time step too large for the speed of pressure waves (a stability limit)	Split each 20 ms step into many small steps
Crash despite a safety check	The check ran every 20 ms; the blow-up happened inside one step	Check the tank speed after every small step
5 mm run crashed at step 100	The wall stayed 4 particles thick, so it became physically thinner	Scale wall layers and time step with particle size
9-hour runs at risk	A shared lab computer can restart mid-run	Save every 25 steps; skip finished runs

Our results in plain words

What each result means, without the jargon

1. **The detailed water simulation can be trusted**
   Made finer and finer, it settles on the exact textbook answer, within about 3%

2. **The quick model gets the rhythm right, but not the lean**
   It predicts how fast the water sloshes (within 6%), but not that the water leans to one side when the tank is held tilted

3. **A controller designed only on the quick model fails on real water**
   LQR lost control in 0.7 s

4. **The learning controller calms real water best**
   15 times less sloshing than PID, but it stays in control for only about 5 seconds

5. **Cheap training, then a short check on detailed physics, works**
   50,000 extra steps made it 3.8× better; more made it worse

Significance of the results

What the results show, and where they can be used

1. **Learning control works on realistic water**
   Not only on the model it was trained on: 15× less sloshing than tuned PID on SPH water

2. **Model-based control can fail when the model is too simple**
   LQR on the pendulum model diverged in 34 steps on real water

3. **A practical design recipe**
   Train cheaply, fine-tune briefly on realistic physics, and stop early

4. **Known limits for the cheap model**
   Trust it for the sloshing frequency (within 6%), not for sustained tilt

5. **For our main application, the road tanker**
   An active tilting cradle can be designed and trained this way, and checked on detailed fluid before it is built

Conclusion

Three questions, three answers

1

Is the detailed water simulation accurate?

Yes, for the slow response: it converges as particles shrink and comes within 3.1% of the exact answer (GCI 6.4%).

2

Can the fast model be trusted?

For the sloshing frequency, yes (within 6%). For a tank held at an angle, no: it misses the water re-levelling.

3

Do the controllers work on real water?

LQR designed on the pendulum fails; PID is stable but leaves more sloshing; PPO leaves ~15× less, but stays stable for only ~5 s.

Limitations

Stated openly, alongside the results

Simulation and data

- No experimental validation: verified against exact answers only
- Only bulk quantities (centre of mass, energy); no wall pressures
- Damping at 10 mm is mostly numerical
- Water viscosity, surface tension and air are not modelled

Control tests

- Two-way runs only at the coarse 10 mm size
- One run per controller
- The learning controller is stable for only ~5 s
- Controllers read the water's position directly, which a real tank cannot easily measure

Future Scope

Next steps from Chapter 6 of our report

1. **Physical validation**
   Build the acrylic pitch rig at this size, measure wall pressure and surface height, then scale up to a tanker compartment

2. **Frequency and damping convergence**
   Repeat free sloshing at 5, 3 and 2 mm to separate real and numerical damping

3. **A better simple model**
   Add the hydrostatic tilt term to the pendulum model and redesign LQR on it

4. **Stronger control tests**
   Repeat the two-way tests at 5 mm with real viscosity, several runs per controller and better walls

5. **Hardware-ready control**
   Use wall-pressure sensors instead of the water's position (a first version was tried)

References

Selected; the full list of 35 references is in the report

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Thank You

Questions are welcome

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Full project on GitHub: https://github.com/RacerRed505/LiquidSloshingRL-Capstone